

**SMART RESEARCH: AN AI-POWERED WEB-BASED RESEARCH PAPER RETRIEVAL
AND RECOMMENDATION SYSTEM USING BERT-NLP**

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INTRODUCTION

The significant volume of higher education institution academic research outputs has resulted in growing large collections of student theses. These papers are invaluable sources of information and reference through which future scholarship may draw upon. Even so, many institutions continue to use conventional archiving systems that are manual or only partly digitized. These systems make information retrieval less efficient because they are mainly dependent on simple keyword searching or physical records maintenance that is often inefficient and error-prone.

To overcome these limitations, The proposed system, Smart Research: An AI-Powered Web-Based Research Paper Retrieval and Recommendation System Using BERT-NLP, aims to provide a more intelligent and convenient way to search, retrieve, and recommend research papers.

By integrating features such as citation count, advanced filtering, and automatic metadata extraction through Optical Character Recognition (OCR), the system ensures accurate and organized retrieval of relevant studies and organized retrieval of relevant studies following the MRAD format.

Background of the Study

in a real-time environment to support researchers in finding and selecting relevant papers easily. In the world of academia, papers and theses are considered a major source of knowledge for both students and professors. However, with the continuous increase in the number of research works, it has become very hard to get the papers needed. The majority of the systems that exist today are still using simple keyword searches, which often give out unrelated or already mentioned results. This very drawback elongates the research cycle and makes it more difficult for the research community to get the studies that are very much related to their topics.

Today, the powerful combination of Artificial Intelligence (AI) and Natural Language Processing (NLP) has allowed intelligent retrieval systems to analyze the user's queries in terms of their intent and context. Among others, BERT (Bidirectional Encoder Representations from Transformers) is the most advanced one that grants the systems a better understanding of natural language which consequently leads to high precision and relevancy of search results. The addition Of BERT to a scholarly retrieval system gives the users not just the recommendations based on the keywords but also through the similarity of research owing to the semantics.

The system that is put forward, Smart Research: An AI-Powered Web-Based Research Paper Retrieval and Recommendation System Using BERT-NLP, has come up with a number of intelligent features for making the research more accessible. Some of them are the monitoring of the citation counts to determine the impact of research, the provision of sophisticated filtering alternatives for more selective searches, and the provision of Optical Character Recognition (OCR) for automatic metadata extraction from the cover pages of uploaded documents. The system provides the functionality of defining the access rights for the roles of administrators, faculty, and users, hence, allowing smooth and secure access control.

The digital repositories, notwithstanding their existence, suffer from the problem that most of them do not possess the AI-backed functions for smart retrieval and recommendation. Thus, we plan to create a web-based platform that will utilize BERT-NLP and OCR in the application of technologies for automating retrieval of research papers and provision of significant recommendations to users is one of the main features of the system. The system intends to use these advanced features to make academic research more accessible, orderly, and efficient for the higher education sector.

Statement of the Problem

There is a substantial collection of student theses and research papers as a result of the ongoing expansion of academic research outputs in higher education institutions. These outputs are frequently kept in conventional paper-based archives or partially digitalized systems, despite the fact that they are extremely valuable as references and information sources for future researchers. Information retrieval is made laborious, time-consuming, and ineffective by such techniques. Reliance on traditional keyword-based search strategies frequently results in irrelevant search results and inadvertently duplicates research topics, which restricts the full use of important scholarly outputs.

The study suggests creating a smart research paper retrieval system that uses abstract-based search and natural language processing (NLP) to overcome these difficulties. The study specifically aims to address the following problems with the way that research papers are currently retrieved:

1. Develop a web-based platform that allows users to search for research papers through natural language queries using BERT-based Natural Language Processing (NLP).
2. Integrate a filtering option and citation count feature to help users refine and evaluate search results based on relevance and scholarly impact.
3. Implement an OCR-based metadata extraction feature for automatic retrieval of research paper details such as title, author, and publication year.
4. Establish user role management with three access levels: Admin, Faculty, and User to ensure data privacy, controlled access, and proper content management.
5. Create a recommendation mechanism that provides relevant research suggestions based on the user's query and previous search patterns.
6. Evaluate the system's performance in terms of accuracy, usability, and efficiency through testing and user feedback.

Project Context

In schools, especially colleges and universities, study papers and theses are important products that show how well students are doing in school and what they have to offer intellectually. However, most colleges continue to use manual or partially digitized archiving systems, making it difficult to find relevant studies quickly. Using simple keyword-based search methods frequently yields irrelevant results, topic redundancy, and time-consuming searches. As a result, great research contributions are neglected, and doing literature reviews becomes tiresome and wasteful for both students and faculty members.

To overcome these challenges, the study suggests creating a Smart Research Paper Retrieval System Using Natural Language Processing (NLP) on Abstract-Based Search. This system uses NLP techniques to comprehend the semantics of user searches and then retrieves the most relevant research publications based on their abstracts. Unlike standard keyword searches, the NLP-based system reads the context and intent of user input, yielding more precise and relevant search results. The system also gives centralized digital access to full-text thesis documents, ensuring that archived research is efficiently kept, retrievable, and protected for future use.

The project's goal is to upgrade the university's research management system by developing an intelligent, automated platform that improves accessibility, lowers redundancy, and increases academic output. This breakthrough enables students to conduct more effective literature reviews, faculty members to direct research with greater precision, and academic institutions to maintain the integrity and preservation of intellectual publications. Finally, this study advances intelligent information retrieval systems by demonstrating the practical application of natural language processing (NLP) in academic archiving and research support.

Objectives of the Study

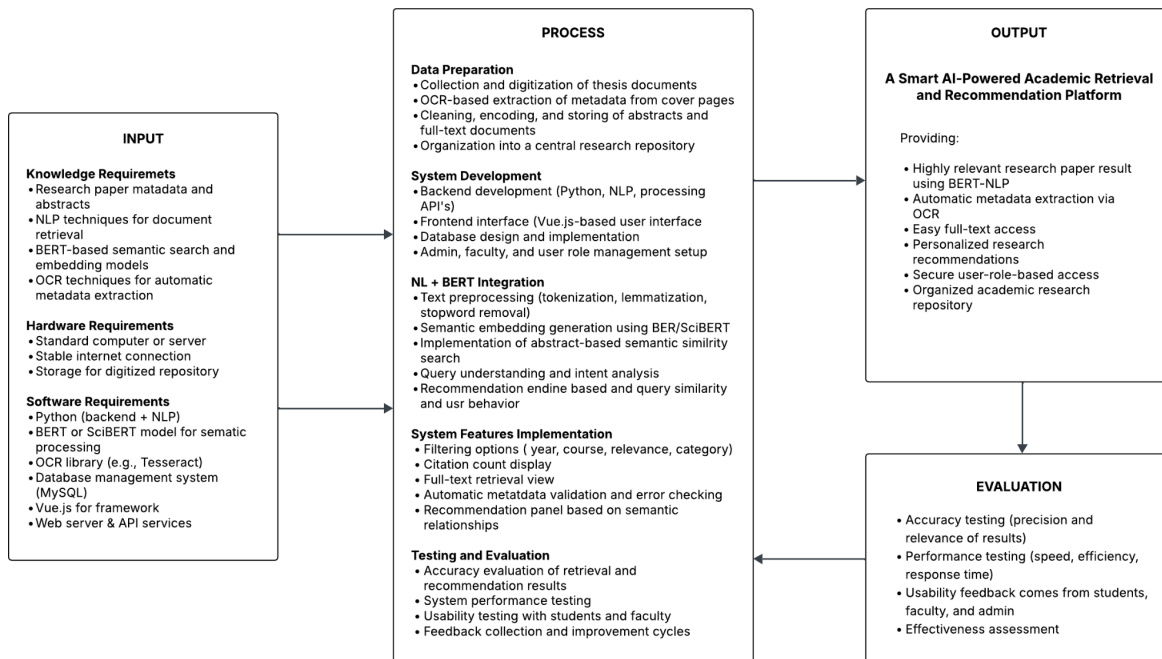
This study aims to develop a smart research paper retrieval system that uses or utilizes the Natural Language Processing (NLP) for abstract-based searches. This system will also give access to full digital copies of past student's research studies.

Specifically, it aims to:

1. Gather, digitize and organize past thesis documents into a central electronic repository.
2. Implement an abstract-based search method with NLP to improve the relevance of search results compared to traditional keyword searches.

3. Allow quick retrieval of full thesis documents once a relevant abstract is found.
4. Help students and faculty conduct literature reviews by offering an efficient and easy-to-use search interface.

Conceptual Framework



Significance of the Study

Its implementation is anticipated to bring great benefits to all interested parties in the education sector in terms of improving the process of carrying out research and modernization in archive management. To Students, the Smart Research system is an indispensable utility in their process of carrying out literature reviews in an improved and faster manner in terms of retrieving relevant studies to their topic or subject using semantic search and saves them considerable time in the initial stages of their literary contributions to society. To Faculty Members and Researchers, improved access to past studies helps in creating awareness about existing gaps and prevents the repetition of topics in the process of carrying out research and offers clearer guidance on future study objectives.

Moreover, the Academic Institutions & Libraries, in particular Cavite State University-Imus, would benefit from having a high-quality digital archiving system which would help in managing student submission collections. Lastly, the outcome of this study could serve as a basis or a stepping stone for Future Researchers as it would create a central, intelligent

space that would encourage continuity in academics, as well as an example of applying Natural Language Processing, BERT, & models in such a setting.

Time and Place of the Study

The study has not been proposed or conducted yet. Once authorized and within the time range established by the organization, it will be executed. During the implementation phase, the precise location and timetable of the study will be decided based on participant and resource availability.

Scope and Limitation of the Study

The goal of the project is to create a Smart Research Paper Retrieval System that improves academic information retrieval by leveraging semantic search methods and Natural Language Processing (NLP). By processing user queries in normal language, the system enables users to look for research papers without depending only on particular keywords. Tokenization, stemming or lemmatization, part-of-speech tagging, and named entity recognition are some of its primary NLP approaches for query processing. Additionally, the system uses semantic understanding to decipher the contextual meaning of user inquiries by utilizing massive language models or word embeddings.

A ranked list of research papers with summaries, abstracts, or direct links to additional reading are produced by the system.

Students, teachers, and researchers that require a quick and dependable tool to support literature reviews and academic research are the main audience for this project. By using semantic understanding rather than keyword reliance, it seeks to increase accuracy, reduce the amount of time spent looking for relevant studies, and improve research retrieval efficiency.

Definition of Terms

Abstract – An abbreviated summary of a research work or a dissertation which highlights its main objectives, methods, findings, and inferences. In this work, abstracts are used as a basis to locate and retrieve relevant documents.

Archiving System – An approach or system used to maintain and preserve records of documents to allow long-term access and recovery. This comparison is about traditional archiving processes as compared to an electronic, intelligent system.

Literature Review – A method of a systematic way of gathering, analyzing, and synthesizing past studies of a particular subject matter. This work seeks to enhance literature review efficiency through an automated information retrieving system.

Natural Language Processing (NLP) – An area of artificial intelligence that allows computers to interpret and understand human language as well as to create human language. This work uses NLP methods to browse user questions and compare them to pertinent abstracts of research articles.

Repository – A central depository facility that is either physical or online whereby papers such as thesis papers and research papers are stored, maintained, and made available.

Retrieval System – An application or platform that is intended to search and retrieve information to reply to a user's query. To enhance accuracy, the intended system implements NLP-based search on an abstract basis.

Intelligent Research Paper Retrieval System – This system suggested in this research utilizes natural language processing techniques to improve the effectiveness and efficiency of retrieving research papers through abstract analysis.

Research paper or thesis – A formal academic document produced by scholars to fulfill their academic responsibility that comprises original work of research. These papers constitute basic content that is to be deposited and retrieved from the provided system.

REVIEW OF RELATED LITERATURE

Foreign Literature

Fundamentals of Natural Language Processing in Information Retrieval Systems

Clarke, Hughes, and Banerjee (2025) investigates the fundamental elements of Natural Language Processing, including tokenization, lemmatization, parsing, and semantic modeling, and underscores their crucial influence on the development of Information Retrieval systems. It likewise examines recent innovations, particularly word embeddings and transformer-based architectures, which have considerably advanced the capacity for semantic interpretation and contextual understanding. At the same time, it acknowledges the persistent challenges that continue to shape the field, such as the complexities of polysemy, synonymy, and multilingual processing. By engaging with these issues and highlighting contemporary research directions, this work aspires to provide a comprehensive and enduring reference for scholars and practitioners, fostering a deeper appreciation of the dynamic interplay between NLP and IR.

Carry out the Library Services and Literature Retrieval Course Using Blended Learning with the Intelligence Services System to Enhance Students' Literature Retrieval Knowledge and Skills

Yuan, Onyon, and Choicharoen (2025) conducted an experiment with 30 first-year students from the School of Literature at Zhoukou Normal University during the second semester of the 2024 academic year, utilizing a cluster random sampling method. To evaluate the effectiveness of the intervention, students were assessed using defined scoring criteria for literature retrieval knowledge and skills, both prior to and following the implementation of the program. The integration of blended learning and intelligent service systems into library services and literature retrieval courses, with the aim of strengthening students' knowledge and skills in literature retrieval and enabling them to master the entire retrieval process. By combining theoretical instruction with hands-on practice, the approach

seeks to cultivate students' interest and motivation in learning, while enhancing both the efficiency and accuracy of their retrieval performance.

Application effect of improved retrieval system based on user understanding on library management system

Li, (July 2025) In the field of library management, improving and optimizing retrieval systems has become a major and urgent issue. As stores of information grow, retrieval systems must be able to accurately understand and meet user needs. But the methods we have now are still limited because they rely too much on surface-level keyword matching and don't get the deeper meaning and context behind a user's question. Not being able to match retrieval processes with the different goals of users makes searches less efficient and makes it harder to reach the larger goal of making information resources easier to find.

Intelligent Semantic Search for Academic Journals Using AI and NLP Techniques

According to Malo, and Al-zebari (2025) the design and implementation of an intelligent semantic search engine specifically designed for academic articles, incorporating Artificial Intelligence (AI) and Natural Language Processing (NLP) methodologies. The system uses sentence transformer models for semantic embeddings to enable vector-based similarity searches, as well as spaCy for tokenization and entity recognition. An ontology-based matching method matches user queries to domain-specific subjects, increasing the precision of search results.

This hybrid framework efficiently combines neural semantic models and structured query formulation, overcoming the constraints of classic keyword-based retrieval systems while increasing research efficiency and access to high-impact literature.

Transition from Traditional Knowledge Retrieval into AI-Powered Knowledge Retrieval in Infrastructure Projects: A Literature Review

According to Rodriguez and Chen (2024) It highlights the shift from traditional knowledge retrieval methods that rely on human rules and statistical models to AI-powered systems that are more flexible and scalable. The authors highlight the limits of traditional retrieval systems, such as their reliance on domain-specific traits and limited adaptability, while contrasting them with AI-powered systems capable of handling complicated data relationships. Their comprehensive research also identifies important obstacles in AI-powered retrieval, such as data bias, processing needs, and comprehension limitations, and proposes alternative methods to address these concerns.

Local Literature

Batayan: A Filipino NLP benchmark for evaluating Large Language Models

Montalan et al. (2025) present Batayan, a complete Filipino benchmark designed to assess large language models (LLMs) in understanding, reasoning, and generating tasks. It focuses on Tagalog and code-switched Taglish, with annotations from local speakers to assure linguistic authenticity and proficiency. The authors highlight the performance disparities of multilingual models on Filipino tasks, emphasizing the language's underrepresentation in global AI datasets, as well as the problems provided by its complex morphology. The study underlines the importance of culturally and linguistically appropriate NLP resources for improving AI applications in the Filipino environment.

Science Mapping of Publications in Natural Language Processing in the Philippines: 2006 to 2020 (2021)

Roxas, Imperial, and De La Cruz (2021) They conducted a comprehensive quantitative and qualitative review of Natural Language Processing (NLP) scientific literature in the Philippines, covering a period of fifteen years from 2006 to 2020. Its aim was to assess the current state of research and identify existing gaps for future improvement. Among the 217 Scopus-indexed papers, 185 were manually screened and evaluated to analyze research productivity, modalities, language polarity, the human languages examined, and the various NLP domains explored. The findings shed light on the status of NLP research in the country and provide recommendations to bolster local research efforts in this area.

Modeling the influence of AI dependence to research productivity among STEM undergraduate students: case of a state university in the Philippines

Buniel, J. M., Juancho Intano, Cuartero, O., Grustan, K. J., & Cortes, S. (2025) Explores the growing dependence on artificial intelligence (AI) among STEM undergraduate students and its relationship to research productivity. It investigates how AI use influences students' research skills, disposition, and self-efficacy, which serve as mediating factors

between AI dependence and research output. Based on survey responses from 834 STEM students, the findings reveal that while AI dependence does not directly affect research productivity, it positively impacts students' research skills, motivation, and confidence. This suggests that the strategic integration of AI resources in academic settings can enhance students' overall research performance and development in STEM fields.

Smart Information Retrieval using Query Transformation based on Ontology and Semantic-Association

According to Kumar, and Harma (May 2022) the shortcomings of modern information retrieval systems, especially how unclear query representations make it difficult for them to accurately identify users' information needs. In order to address this issue, the researchers created an ontology-based retrieval system that uses domain-specific ontologies to convert user queries into more insightful semantic representations. The system improves the correspondence between indexed terms and user queries by using semantic associations in the indexing process. According to evaluation results, retrieval accuracy and efficiency are significantly increased when ontology and semantic matching are combined, as opposed to using traditional methods. This highlights the importance of semantic understanding in contemporary information retrieval.

Development of a Web-based records management system: an ERMS initiative for the Office of Senior Citizen Affairs in the Philippines

Botangen, Libadia, Lucero, and Tecson (2025) studied and focused on creating a web-based Electronic Records Management System (ERMS) for the Philippine Office of Senior Citizens Affairs (OSCA). In order to overcome the inefficiencies of manual record management, the system seeks to centralize record-keeping and expedite the gathering, storing, and processing of member information. The ERMS improves the speed and accuracy of information retrieval while maintaining data security and privacy through the use of web and data-capturing technologies. The system's usability, dependability, and efficiency

are confirmed by an evaluation using the ISO 25010 product quality model. The project shows how web technologies and structured data management can be integrated to create locally developed systems that optimize government office workflows.

REVIEW OF RELATED STUDIES

Foreign Studies

Document Retrieval Using Deep Learning

Choudhary, S. et al. (2020) The study addresses the limitations of traditional document retrieval systems, which often rely on bag-of-word approaches that fail to capture the semantic context necessary for understanding document-query relevancy. The research explores the integration of advanced Natural Language Processing techniques, specifically leveraging Bidirectional Encoder Representations from Transformers (BERT) to create semantically rich document embeddings. By proposing an ensemble model that combines BERT with the traditional Term Frequency-Inverse Document Frequency (TF-IDF) method, the authors aim to improve the precision of document scoring against user queries. Their investigations, conducted using the MS MARCO dataset, demonstrate that this hybrid approach significantly outperforms standard TF-IDF models by incorporating contextual nuances that fixed-length document representations typically overlook. The broad implication of these findings is that combining deep learning-based contextual embeddings with traditional statistical methods provides a more robust and effective framework for modern information retrieval systems.

Neural ranking models for document retrieval

Mohamed Trabelsi, Zhiyu Chen, Brian D. Davison, Jeff Heflin (2021) In the past, ranking strategies have depended on machine learning algorithms that use human designed criteria to assess the relevance of documents. However, these approaches reliance on preset features frequently limits them. This method has been completely changed by recent developments in deep learning, which allow models to learn feature representations from raw data in an end-to-end fashion. This change enables deep learning-based ranking models to overcome the drawbacks of conventional methods, producing retrieval results that are more precise and flexible. Different neural architectures with unique methods for

extracting pertinent characteristics have been presented for rating problems. The advantages and disadvantages of these deep ranking models are emphasized by comparative evaluations in related studies, which also offer insights into the brain components that work well and suggest areas for further research. Deep learning's versatility and promise to improve intelligent information retrieval systems, including research paper search platforms, are further highlighted by the similarities between document retrieval and other structured retrieval challenges, such as answer, picture, and video retrieval.

Improving Embedding-based Large-scale Retrieval via Label Enhancement

Peiyang Liu, Xi Wang, Sen Wang, Wei Ye, Xiangyu Xi, and Shikun Zhang (2020)

Binary labels (0 or 1) are commonly used in the training of traditional embedding-based large-scale retrieval models to signal whether a document is pertinent to a particular query. This method, however, ignores the different levels of relevance that can exist between documents and queries. A recent study addressed this constraint by introducing a Label Enhancement (LE) technique that aims to better capture the degree of relevance in order to improve the representation of query–document interactions. The suggested approach incorporates past information from dynamic word weighting approaches with contextual embeddings to provide a label distribution for retrieval tasks. During model training, the improved labeling technique offers fuller supervision, which raises retrieval accuracy. The LE-based model beat four leading retrieval baselines and alternative label enhancement techniques under both standard and cold-start conditions, according to experimental assessments conducted on large-scale English and Chinese datasets. This work emphasizes how crucial it is to include subtle relevance signals in embedding-based retrieval systems. This strategy can be especially helpful for intelligent research paper retrieval applications that make use of natural language processing (NLP) approaches.

Cognition based Smart Document Matching using Chatbot Technology

Harsh Tomar, Ravi Sharma, Sahil Prakash (2025) The ability to facilitate smooth natural language communication between humans and machines is one of the core objectives of artificial intelligence (AI). In order to enhance the response to human resource (HR) queries, a recent study presented a hybrid document retrieval chatbot system that blends Natural Language Processing (NLP) with Machine Learning (ML) approaches. The system uses a combination of Sentence-BERT embeddings and Term Frequency–Inverse Document Frequency (TF-IDF) to represent both documents and user queries in a shared vector space. Cosine similarity is then used to rate document relevance, and internal testing shows that this method can achieve up to 92% top-3 retrieval accuracy. To further reduce retrieval time by roughly 25%, a logistic regression-based intent classification module was put in place to group user queries into particular HR-related domains (such as leave, payroll, and corporate policies). The integration of neural and symbolic approaches can result in more accurate and efficient domain-specific information retrieval, as demonstrated by this modular and scalable design. This strategy can also be used in intelligent research paper retrieval systems that use natural language processing (NLP) for abstract-based searching.

Transformer based contextual text representation framework for intelligent information retrieval

Amol P. Bhopale, Ashish Tiwari (2024) To determine if a document is pertinent to a particular query, binary labels (0 or 1) are commonly used in the training of traditional embedding-based large-scale retrieval models. This method, however, ignores the different levels of relevance that can exist between documents and queries. A recent study addressed this constraint by introducing a Label Enhancement (LE) technique that aims to better capture the degree of relevance in order to improve the representation of query–document interactions. By incorporating past information from dynamic word

weighting techniques into contextual embeddings, the suggested approach creates a label distribution for retrieval tasks. Greater monitoring during model training is made possible by this better labeling technique, which raises retrieval accuracy. The LE-based model beat four leading retrieval baselines and alternative label enhancement techniques under both standard and cold-start conditions, according to experimental assessments conducted on large-scale English and Chinese datasets. This work emphasizes how crucial it is to include subtle relevance signals in embedding-based retrieval systems. This strategy can be especially helpful for intelligent research paper retrieval applications that make use of natural language processing (NLP) approaches.

Local Studies

Developing BERT-based ensemble for Filipino text classification

Ong, L. E. C., & De Leon, M. M. (2025) The study focuses on the critical need for advanced automated content moderation within the context of Filipino social media, where linguistic nuances and the low-resource nature of the language present significant challenges for standard classification models. The research investigates the development of a BERT-based ensemble approach to improve the detection of Filipino hate speech, moving beyond the limitations of single-classifier systems. By exploring various ensemble strategies such as hard voting, soft voting, blending, and stacking, the authors demonstrate how combining multiple fine-tuned BERT models can achieve superior accuracy and robustness. The investigations specifically highlight that meta-classification techniques like stacking are particularly effective at capturing complex linguistic patterns in Filipino text. Ultimately, the findings suggest that implementing these sophisticated ensemble models provides a more reliable framework for identifying offensive content, offering essential guidance for developers and researchers aiming to foster safer digital environments in the Philippines.

Open Law Philippines: Legal Document Retrieval Analysis

A., Licup, P., Villarama, K., Permito, J., Laguna, A., & Robles, D. M. (2024) In creating a document retrieval system especially for Philippine law, the project aims to overcome the difficulty of organizing and analyzing the country's vast legal document collection. In order to increase the precision of legal text analysis and semantic document search, the research incorporates cutting-edge deep learning techniques and natural language processing (NLP) models like Juris2Vec, LEGAL-BERT, and Sentence-BERT (SBERT), taking inspiration from projects like Harvard's Case Law Project. The system seeks to improve the accessibility and understandability of legal information for practitioners, scholars, and the general public by partnering with legal professionals and integrating BERTopic for theme categorization. This initiative encourages creativity and accessibility in

legal research and documentation by becoming the first to locally apply cutting-edge AI models in Philippine legal informatics.

The Influence of Artificial Intelligence (AI) on Library Patronage among College Students

Soberano, L., Reyes, R., Fabro, R., (2025) Study the rising trend among students is to utilize AI instruments for prompt information retrieval and study suggestions due to their swiftness, ease of use, and round-the-clock accessibility. Nevertheless, studies also show that academic libraries still hold significant importance, notably when it comes to accessing dependable databases and carrying out thorough investigations. Scholarly works underscore that students perceive both AI and library resources as working together, not acting as substitutes for one another. In light of this, academics propose combined frameworks that incorporate AI alongside established library offerings, accompanied by instruction in digital understanding and ethical standards for AI implementation.

Web-Based Online Platform for Research Study

Delera, G., Salao, M., & Soberano, K. (2023) Investigates the necessity of digital research repository systems that offer dependable, well-managed, and protected storage for scholarly papers. The existing body of knowledge also calls attention to the significance of building accessible platforms that adhere to the criteria set by institutions to guarantee straightforward implementation. A lot of system creation research employs Agile-centered SDLC models, given that progressive creation supports the improvement of system attributes according to what users need. Frequently, research projects assess these platforms via questionnaires given to research administrators and students, indicating substantial approval and enthusiasm stemming from the improved handling and availability of research resources. Taken as a whole, the available research indicates that repository systems successfully maintain and structure digital scholastic outputs.

Evaluation of Web Elements of Selected Philippine Academic Library Websites

Tamayo, M. C., Factor, M., & Surio, J. R. (2024) The study highlights the COVID-19 global crisis that played a major role in expediting the transition of libraries to online accessibility, compelling these institutions to make their resources, offerings, and initiatives available using digital means. Investigations highlight that progress in technology has reformed the way people get information and increased how vital well-thought-out library websites are when it comes to backing up academic demands. Studies further accentuate how critical precise guidance is in enabling library experts and designers to create or refine these websites. Certain research projects implement assessment tools, for example, the system created by Wijayaratne and Singh in 2015, to judge the value of academic library websites and find spots that need work. The broad implication of these findings is that it's important to construct customized and user-focused library websites that successfully handle the demands of learners, instructors, and scholars

Research Gap

Analysis of the reviewed studies shows that there is an enormous technological gap in research management implementation in comparison to global developments. Across the world, studies such as “Cognition based Smart Document Matching using Chatbot Technology” (2025) and “Document Retrieval Using Deep Learning” (2020) have already reached an incredible milestone in BERT-based semantic searching by proving that AI can interpret the meaningful context of the researcher’s query much better than the traditional searching method. On the contrary, these studies overlook the crucial step of converting physical records to digital data.

Locally, the current situation, as exemplified by research such as Web-Based Online Platform for Research Study (2023), reveals that although there is a shift toward digitalization among Philippine institutions of learning, there is a tremendous reliance on human-data entrying and simple web-based repositories. Indeed, these tools function merely as passive repositories rather than active research assistants. Notably, although there is a focus on innovative NLP, such as in the "BERT-based ensemble for Filipino text classification" research of (2024), it has not been used for a retrieval-and-recommendation system.

However, the major research gap that has been identified is that no such local platform has been developed which can resultantly combine Optical Character Recognition Techniques for the Automated Digitization of Hard-Copy Thesis Archives into BERT-NLP for Context-Aware Semantic Searches as well as Similarity-based Recommendations. At present, no local system has an Automated Mechanism for Monitoring the Citations Count Tracking that is a major factor for evaluating Research Impact Outputs for institutions. Your research work entitled “Smart Research: An AI-Powered Web-Based Research Paper Retrieval and Recommendation System” connects all major research gaps by providing a smarter environment for Philippine universities to conduct their research efficiently.

Synthesis (Table of Comparison)

Table 1. Table of Comparison for foreign studies

	DRUDL	IEBLRLE	CBSDMCT	TBCTRF	NRMDR
Year	2020	2021	2025	2024	2021
Methodology	BERT & TF-IDF	Label Enhancement	S-BERT & TF-IDF	Transformer	Neural Ranking
Features:	-	-	-	-	-
NLP-Based Semantic Search	✓	✓	✓	✓	✓
Abstract-Based Retrieval	✓	X	X	✓	X
OCR Metadata Extraction	X	X	X	X	X
Recommendation Mechanism	X	✓	✓	X	X
Citation Count Tracking	X	X	X	X	X
User Role Management	X	X	✓	X	X

LEGEND:

DRUDL - Document Retrieval Using Deep Learning

IEBLRLE - Improving Embedding-based Large-scale Retrieval via Label Enhancement

CBSDMCT - Cognition based Smart Document Matching using Chatbot Technology.

TBCTR - Transformer based contextual text representation framework for intelligent information retrieval

NRMDR - Neural ranking models for document retrieval.

Table 2. Table of Comparison for local studies

	DBEFTC	WBOPRS	IAIL-P	EWES-PAL	SRAIPWBR PRRSUB
Year	2024	2023	2025	2024	2026
Methodology	BERT Ensemble	Web development	AI Framework	Web Evaluation	BERT-NLP & OCR
Features:	-	-	-	-	-
NLP-Based Semantic Search	✓	X	✓	X	✓
Abstract-Based Retrieval	X	X	X	X	✓
OCR Metadata Extraction	X	X	X	X	✓
Recommendatio n Mechanism	X	X	✓	X	✓
Citation Count Tracking	X	X	X	X	✓
User Role Management	X	✓	X	X	✓

LEGEND:

DBEFTC - Developing BERT-based ensemble for Filipino text classification

WBOPRS - Web-Based Online Platform for Research Study

IAILP - The Influence of Artificial Intelligence (AI) on Library Patronage among College Students

EWESPAL - Evaluation of Web Elements of Selected Philippine Academic Library Websites

SRAIPWBRPRRSUB - Smart Research: An AI-Powered Web-Based Research Paper Retrieval and Recommendation System Using BERT-NLP

METHODOLOGY

This chapter outlines the proposed method for developing and evaluating the proposed system. The procedures described here represent the planned way to meet the study's goals through an iterative development process.

Research Design

The researchers will use a Developmental Research Design. This strategy offers a clear framework for designing, developing, and evaluating the web-based retrieval system, ensuring that the software effectively tackles the issue of ineffective research paper retrieval.

Research Approach

The study will take a mixed-methods approach to assess the system's performance and usefulness. The quantitative part will gather numerical data about the system's technical accuracy, especially the precision and relevance of the BERT-NLP search results, along with performance metrics like response time. The qualitative part will include collecting descriptive feedback and suggestions from users during the development stages to refine the system's interface and recommendation features.

Research Settings

The study will take place at Cavite State University, Imus Campus. This location is chosen as the main source of research data and the environment for implementing the system.

Participants / Respondents / Data Sources

- **Data Source:** The primary data will come from the university's repository of undergraduate thesis documents, focusing on metadata and abstracts.

- **Respondents:** A targeted sample of students and faculty members from the Department of Computer Science will evaluate the system as the main end-users.

Research Instruments

The researchers will develop a set of tools to capture both technical and user data. A Software Quality Evaluation Form will gather structured feedback on the system's functionality and ease of use, allowing respondents to rate their experience with numbers. To assess the intelligence of the platform, an Accuracy Testing Log will record how accurately the BERT-NLP model retrieves relevant documents based on user intent. Finally, a User Feedback Questionnaire will be given to collect qualitative insights, ensuring that specific suggestions for the recommendation engine and interface are noted for future updates.

Data Collection Procedure

The planned procedure consists of three main stages. It begins with Data Gathering, where past thesis papers are collected and digitized into a central electronic repository. Next comes Extraction and Cleaning, where the team will use OCR to extract metadata and clean the text for AI processing. The final stage is User Testing, involving controlled sessions that let respondents interact with the system and give feedback for improvement.

Data Analysis Techniques

To interpret the results, the researchers will use the Statistical Mean to analyze user satisfaction and usability ratings from the questionnaires. For the technical side, Information Retrieval Metrics, especially precision and recall, will evaluate the accuracy and relevance of the BERT-based similarity search engine compared to traditional keyword searches.

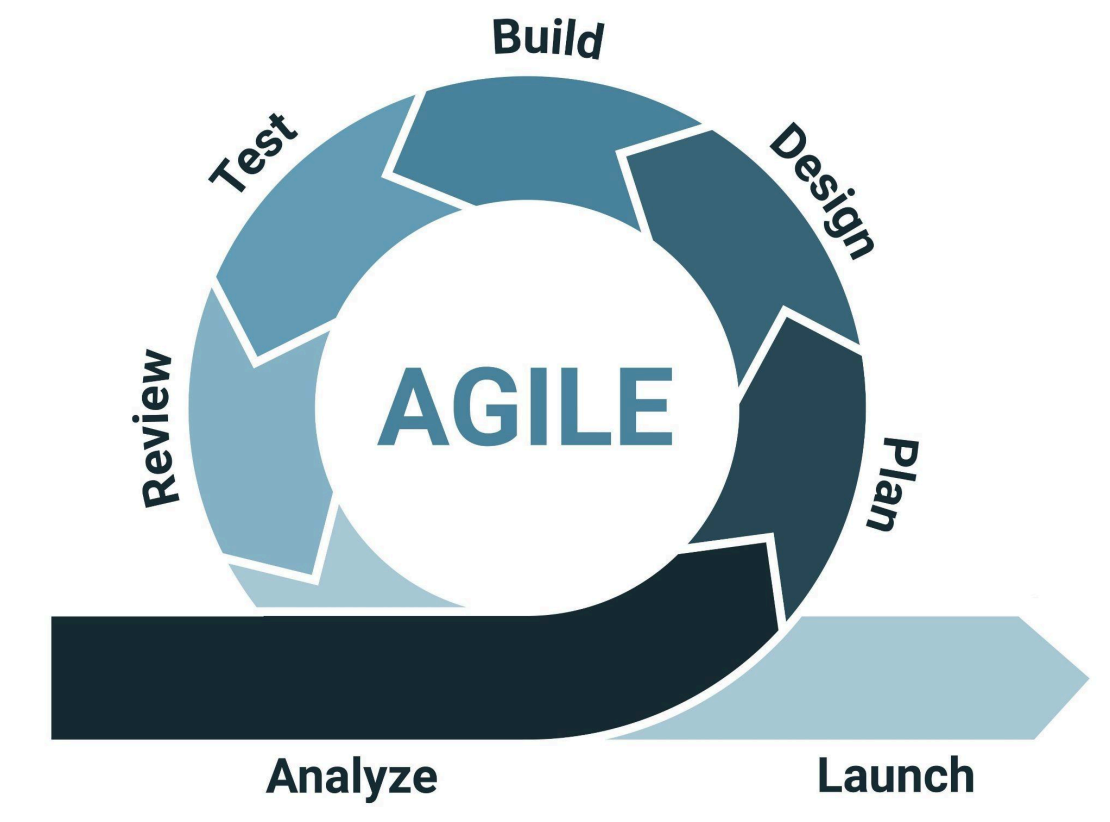
Ethical Considerations

The researchers will put safeguards in place to protect Intellectual Property by limiting full-text access according to university guidelines. They will maintain confidentiality

for all participants in the testing phase, and Informed Consent will be obtained before any evaluation sessions to ensure that participation is voluntary and clear.

System Architecture / Development Model

The proposed study and the system will be developed using the Agile Iterative Model. This approach allows the researchers to continuously improve the system's intelligent features through ongoing cycles of planning, development, and testing. The project will be broken into Sprints, which will enable the researchers to incrementally build the OCR metadata extraction and the BERT-based semantic search while making real-time adjustments based on each iteration's results.



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